

Thermo Scientific™ DynaDrive™ Single-use Bioreactor 500 L - 25-500 L working volume

thermo-dynadrive-500

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	25 L	Source-supported
Maximum working volume	500 L	Source-supported
Nominal vessel / bag size	500 L	Source-supported
Total vessel volume loaded into the form	586 L	Source-supported
Initial working volume loaded into the form	25 L	Derived / estimated
Product-contact material	Plastic / single-use	App assumption
Height : diameter	2.5	Source-supported
Internal / chamber diameter	0.634 m	Source-supported
Liquid height at maximum working volume	1.59 m	Source-supported
Impeller / mixing type	Hydrofoil	App assumption
Impeller count	4	Source-supported
Impeller diameter	0.229 m	Source-supported
Power number	0.4	App assumption
Initial agitation	80 rpm	App assumption
Maximum agitation	120 rpm	Source-supported
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	No	App assumption
Maximum pressure	0.03 bar(g)	App assumption
Default operating pressure	0 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 25 L

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Table 4.3, printed page 79

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 25 L.

Limitations: Chamber diameter and liquid height copied directly. Arrangement is three modified pitched-blade plus one sweep impeller; generic single-type modelling cannot reproduce this exactly.

Local original: [Thermo DynaDrive 50 and 500 L user guide - PDF page 83](#)

[Manufacturer publication](#)

Maximum working volume - 500 L

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Table 4.3, printed page 79

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 500 L.

Limitations: Chamber diameter and liquid height copied directly. Arrangement is three modified pitched-blade plus one sweep impeller; generic single-type modelling cannot reproduce this exactly.

Local original: [Thermo DynaDrive 50 and 500 L user guide - PDF page 83](#)

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Nominal vessel / bag size - 500 L

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Table 4.3, printed page 79

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 500 L.

Limitations: Chamber diameter and liquid height copied directly. Arrangement is three modified pitched-blade plus one sweep impeller; generic single-type modelling cannot reproduce this exactly.

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Total vessel volume loaded into the form - 586 L

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Table 4.3, printed page 79

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 586 L.

Limitations: Chamber diameter and liquid height copied directly. Arrangement is three modified pitched-blade plus one sweep impeller; generic single-type modelling cannot reproduce this exactly.

Local original: [Thermo DynaDrive 50 and 500 L user guide - PDF page 83](#)

[Manufacturer publication](#)

Initial working volume loaded into the form - 25 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(25 L, 500 L) = 25 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 2.5

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Table 4.3, printed page 79

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.5.

Limitations: Chamber diameter and liquid height copied directly. Arrangement is three modified pitched-blade plus one sweep impeller; generic single-type modelling cannot reproduce this exactly.

Local original: [Thermo DynaDrive 50 and 500 L user guide - PDF page 83](#)

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Internal / chamber diameter - 0.634 m

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Table 4.3, printed page 79

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.634 m.

Limitations: Chamber diameter and liquid height copied directly. Arrangement is three modified pitched-blade plus one sweep impeller; generic single-type modelling cannot reproduce this exactly.

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Liquid height at maximum working volume - 1.59 m

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Table 4.3, printed page 79

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.59 m.

Limitations: Chamber diameter and liquid height copied directly. Arrangement is three modified pitched-blade plus one sweep impeller; generic single-type modelling cannot reproduce this exactly.

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Impeller / mixing type - Hydrofoil

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].impellerType = "hydrofoil"`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. The selected type determines the generic mixing archetype shown and is paired with the preset power-number assumption.

Impeller count - 4

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Table 4.3, printed page 79

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 4.

Limitations: Chamber diameter and liquid height copied directly. Arrangement is three modified pitched-blade plus one sweep impeller; generic single-type modelling cannot reproduce this exactly.

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Impeller diameter - 0.229 m

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Table 4.3, printed page 79

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.229 m.

Limitations: Chamber diameter and liquid height copied directly. Arrangement is three modified pitched-blade plus one sweep impeller; generic single-type modelling cannot reproduce this exactly.

Local original: [Thermo DynaDrive 50 and 500 L user guide - PDF page 83](#)

[Manufacturer publication](#)

Power number - 0.4

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].powerNumber = 0.4`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. The power number is an empirical generic value used in $P = N_p N_3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 80 rpm

App assumption | Large mammalian-vessel scaling rule

Rule: `baseRpm = maxVolume >= 1000 L ? 55 : 80`

Selection / calculation: 500 L is not at least 1000 L -> 80 rpm

Limitations: The simulator lowers its generic starting agitation at larger scale. This is an internal scale heuristic, not a manufacturer limit. This is the initial agitation setting loaded into the form.

Maximum agitation - 120 rpm

Source-supported | Thermo DynaDrive 50 and 500 L user guide - PDF page 83; Agitation VFD range

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 120 rpm.

Limitations: 35-120 rpm; recommended minimum for culture is above 50 rpm.

Local original: [Thermo DynaDrive 50 and 500 L user guide - PDF page 83](#)

Manufacturer publication

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxVvm = 0.3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - No

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baffled = false`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.03 bar(g)

App assumption | Thermo DynaDrive 50 and 500 L user guide - PDF page 8; BPC operating limits; compare PDF p. 6 burst warning

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0.03 bar(g).

Limitations: Conflict: PDF p. 6 states 34 mbar, while p. 8 states 30 mbar; both print 0.5 psi. The simulator selects the lower 0.030 bar ceiling pending clarification. This is a conservative choice, not a resolved manufacturer rating. Inlet and jacket limits are distinct.

Local original: [Thermo DynaDrive 50 and 500 L user guide - PDF page 8](#)

Manufacturer publication

Default operating pressure - 0 bar(g)

App assumption | Thermo DynaDrive 50 and 500 L user guide - PDF page 8; BPC operating limits

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0 bar(g).

Limitations: Atmospheric app starting value replaces 0.05 bar, above both published BPC limits. Not a manufacturer process setpoint.

Local original: [Thermo DynaDrive 50 and 500 L user guide - PDF page 8](#)

[Manufacturer publication](#)

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

DynaDrive is represented with a generic mammalian stirred-tank profile unless model-specific values are explicitly stored.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Thermo DynaDrive 50 and 500 L user guide

[thermo-dynadrives-50-and-500-l-user-guide-5c184cd19fa7.pdf](#)

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